

DIGITAL GHG INVENTORY SYSTEM

Enterprise Greenhouse Gas Calculations & Reporting

JSW STEEL CO-OP PRESENTATION

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Core infrastructure, database configuration, and libraries used to power calculations:

Layer	Technologies Used	Purpose & Role in Project
Database	PostgreSQL, SQLAlchemy ORM, Alembic	Secure relational data storage for sheet values, sites, users, and audit logs. Schema versions are maintained dynamically via Alembic migrations.
Backend	Python 3.12, Flask, simpleeval	Core API and routing framework. The simpleeval package serves as the runtime mathematical evaluation engine for complex, user-defined GHG formulas.
Frontend	HTML5, Vanilla CSS, Tailwind CSS, Javascript (ES6+)	Interactive spreadsheet UI, auto-saving annual workbook, visual form builder, review route workflow canvas, and real-time dashboard feedback.
Utility & Security	openpyxl, reportlab, bcrypt	Parses/downloads Microsoft Excel templates, builds presentation reports in PDF format, and handles enterprise-grade password encryption.

Key modules responsible for managing form configurations, data inputs, and time frames:

1. Annual Workbook Submission Module (SUBMIT)

- Provides an excel-like grid interface where Site Power Users (SPOCs) enter monthly activity data.
- Implements client-side calculations, auto-save drafts, locking indicators, and cell comment reviews.
- Displays a clear color-coded MONTH column status legend and highlights the active period row with a deep navy border.

2. Interactive Form Builder Module (FORMBLD)

- Empowers administrators to dynamically build sheet templates with custom headings, sections, and input types.
- Supports drag-and-drop ordering, validation limits, custom units, and mandatory attachment flags.

3. Metadata Helpers: Value Sets (VALSET) & Period Management (PERIOD)

- **VALSET:** Manages reusable database dictionaries and dropdown select options (e.g. fuel types, material sets).
- **PERIOD:** Controls active financial years (April to March schedule) and locking/unlocking monthly reporting windows.

Modules handling mathematical transformations, compliance auditing, and executive reports:

1. Emission Formula Runtime Engine (FRMULA)

- Decouples mathematical logic from code; developers or admins write standard equations directly in the builder.
- Parses and evaluates formulas (e.g., *Activity Data* × *Emission Factor* × *GWP*) in a sandboxed simpleeval environment.
- Performs calculation chain resolution, displaying real-time calculation previews alongside final reportable approved values.

2. Report Builder & Aggregations (RPTBLD)

- Aggregates emissions data across sites, Scopes (1, 2, 3), and source categories (stationary, mobile, process).
- Exports compliant spreadsheets and structured summary reports for sustainability officers and statutory audits.

3. Audit Logging & Compliance (AUDITL)

- Logs every user action including value entry, file uploads, approval reviews, comments raised, and state transitions.
- Creates an immutable historical log ensuring full auditability of JSW's carbon footprint disclosures.

Modules managing review processes, permissions, and security controls:

1. Workflow Builder Canvas (WFLWBLD)

- Enables custom approval paths per site, mapping reporting steps to designated reviewers and final approvers.
- Integrates validation checks requiring all sheets to be fully approved before final financial year submission.

2. Reviewer & Approver Queue Dashboard (APPROV)

- Central dashboard showing pending submissions, sheets needing correction, and completed reviews.
- Allows review personnel to view exact discrepancies, post remarks, attach verification stamps, or send sheets back for corrections.

3. Authentication & Role-Based Access Control (ACCESS & USRMGMT)

- **ACCESS:** Implements secure endpoint authorization ensuring SPOCs, Reviewers, and Approvers only access permitted domains.
- **USRMGMT:** Manages employee database directories, site boundaries, and cryptographically hashed passwords.

Integration and system-wide messaging modules:

1. Event-Driven Notification Engine (NOTIFY)

- Automatically triggers real-time emails and in-app alerts on critical workflow changes.
- Alerts SPOCs immediately when a sheet is sent back (rejection/correction requested) to minimize reporting delays.
- Reminds reviewers of pending tasks to ensure reporting schedule compliance.

2. Site Master & Configuration Registry (SITEMST)

- Defines physical plant facilities (e.g. JSW Vijayanagar, Dolvi, Salem) and coordinates their emission boundaries.
- Maps template workbooks to sites dynamically depending on site operations (steel production vs cement vs power plants).

Conclusion

The Digital GHG Inventory System successfully modularizes data collection, validation, execution, and reporting, establishing a robust and transparent sustainability dashboard for JSW Steel.